19 There are only blue cubes, yellow cubes and green cubes in a bag.

There are

twice as many blue cubes as yellow cubes
and four times as many green cubes as blue cubes.

Hannah takes at random a cube from the bag.

Work out the probability that Hannah takes a yellow cube.

For every 1 yellow cube, there are 2 blue cubes.

1 mark
Use the first relationship

Green cubes: $4 \times 2 = 8$ for every 1 yellow cube.

Blue	Yellow	Green
2	1	8

1 mark
Combine both relationships

Total = 2 + 1 + 8 = 11 equal parts.

$P(\text{yellow}) = \frac{1}{11}$

1 mark
Yellow share of the whole

$\frac{1}{11}$

(Total for Question 19 is 3 marks)